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Summary

Caltech PhD student in Economics working across machine learning, empirical asset pricing, and economic theory. Build end-to-end research pipelines from large-scale alternative data collection. Current projects use NLP and large language models to predict UK patent-litigation outcomes and alternative data to study equity-return predictability.

Education

- 2021–2027 **Ph.D. in Quantitative Economics and Econometrics**, *California Institute of Technology* (exp.) (*Caltech*), Pasadena, CA, USA
Selected coursework: Machine Learning, Deep Learning, Advanced Econometrics
- 2017–2020 **M.Sc. in Economics and Social Sciences**, *Bocconi University*, Milan, Italy
Bocconi Graduate Merit Award
- 2012–2016 **B.A. in Business and Management**, *National Economics University of Vietnam*
Merit awards for performance in the top 5% of students (2013, 2014, 2015)

Selected Research

Predicting UK Patent-Validity Court Outcomes

Job Market Paper

- Built NLP models to predict patent invalidation in UK courts—an event associated with an average 1.4% decline in the patent owner's stock price and a potential source of trading alpha.
- Investigated how predictive performance changes between pre-hearing and post-hearing stages, which legal and patent documents contribute most to prediction, and whether separate or joint text encoding performs better.
- Constructed an original dataset spanning 30 years of first-instance UK patent-validity litigation using web scraping and LLM-assisted extraction from fragmented legal and patent records.
- Developed a new patent-text representation and a custom neural architecture to jointly incorporate information from multiple documents.
- Outperformed zero-shot ChatGPT across all prediction tasks on held-out UK court decisions from 2022–2026, for example achieving a macro-F1 score of 0.656 and ROC-AUC of 0.743 on one task.

Do Reviews Affect Stock Performance? Evidence from the Entertainment Industry

Work in progress

- Studied whether critic and user reviews predict stock returns for Japanese and U.S. video game companies.
- Built Scrapy pipelines to collect critic and user reviews from Metacritic and used Python, pandas, and SQL to construct daily review-based signals aligned with financial-market data.
- An event-driven portfolio strategy generated about 11.29% annualized Fama–French five-factor alpha in the study sample, with a t-statistic of 3.91, a Sharpe ratio of 1.1, and a maximum drawdown of around 12.52%.
- The strategy remained robust across alternative holding horizons, rebalancing frequencies, and signal thresholds.

Optimal Stopping Problems with Multiple Agents

Single-authored; accepted for poster presentation at the ACM Conference on Economics and Computation (EC)

- Studied an optimal stopping problem without transfers, in which a principal with state-dependent payoffs solicits sequential reports from multiple privately informed experts who prefer earlier stopping.
- Characterized the optimal stopping rule: the principal commits to a deadline at which she stops with probability one. Before the deadline, she stops with probability strictly between zero and one if and only if the number of high-signal reports exceeds a threshold.
- Showed that the optimal information-aggregation mechanism asymptotically achieves full efficiency. The model applies to an investor timing an asset purchase using advice from analysts paid a fixed commission upon purchase.

Research and Teaching Experience

Teaching Assistant — California Institute of Technology (Caltech)

- Teaching assistant for *Networks and Structure in Economics*, *A History of Budgetary Politics in the United States*, and *Understanding China Through Finance*; designed and graded problem sets and exams, led discussion sections, and delivered guest lectures.

Research Assistant — Bocconi University

Market Microstructure (Dark Pools)

- Built and analyzed a game-theoretic model in Mathematica to study the introduction of dark pools, showing that their addition increases market depth and reduces spreads.
- Specified model primitives and the equilibrium concept, implemented equilibrium computation, and conducted comparative analyses across market structures.

Technical Skills

Programming	Python (PyTorch, TensorFlow, pandas, NumPy, scikit-learn, statsmodels, Scrapy, Playwright, Transformers), SQL, DuckDB
Machine learning	Supervised learning, neural networks, NLP/LLMs, pretrained embeddings, fine-tuning and LoRA, classical ML methods (logistic regression, SVMs, tree-based models), and model ensembling
Data engineering	Built large-scale web scraping and data-processing pipelines using Scrapy, Playwright, pandas, NumPy, SQL, and parallel processing; worked with both structured data and unstructured text
Empirical methods	Event studies, portfolio backtesting, econometrics, feature engineering
Theory & optimization	Dynamic programming, convex optimization, integer programming, game theory, mechanism design